

Data Cleaning & Preparation — Change Log

Project 1 · Data Analytics Internship · Decode Labs

Intern Name:	Samson Oluwatimilehin Arawande	Department:	Data Analytics
Position:	Data Analytics Intern	Mode:	Remote / Virtual
Internship Period:	27 Sep 2026 – 27 Oct 2026	Report Date:	02 Oct 2026
Dataset:	Dataset_for_Data_Analytics.xlsx	Tool Used:	Microsoft Excel (Power Query)

Objective

Clean the raw dataset by identifying and handling missing values, removing duplicate records, and correcting inconsistent data formats (dates, numbers, text), in line with the Project 1 brief issued by Decode Labs.

Dataset Summary

Metric	Value
Total records (rows)	1,200
Total fields (columns)	14
Primary key identified	OrderID
Distinct OrderID count	1,200
Duplicate OrderID count	0
Full-row duplicate count (all 14 columns)	0

Change Log

Change ID	Description	Impact	Status
CR001	Rounded UnitPrice and TotalPrice to 2 decimal places using Power Query (Transform → Rounding → Round).	Removed floating-point rounding artifacts (e.g. 769.37999999999982 → 769.38) on 29 rows; standardized numeric precision across all 1,200 rows.	Resolved
CR002	Filled 309 blank CouponCode values with the explicit label "No Coupon".	Removed ambiguity between missing data and intentional non-use of a coupon; dataset now has zero blank cells in this field.	Resolved

Change ID	Description	Impact	Status
CR003	Converted the Date column from Date/Time to Date-only, then formatted as ISO 8601 (YYYY-MM-DD).	Removed redundant 12:00 AM timestamps present on all 1,200 rows; standardized date format for consistency and reporting.	Resolved
CR004	Validated OrderID as the dataset's primary key via a Power Query Group By check (OrderID, Count Rows) and Excel formula verification (ROWS, COUNTIF).	Confirmed 1,200 total rows and 1,200 distinct OrderID values; 0 records with a duplicate OrderID.	Resolved
CR005	Applied Trim and Clean (Transform → Format) to all 6 text columns — Product, ShippingAddress, PaymentMethod, OrderStatus, CouponCode, ReferralSource.	Safeguarded against stray leading/trailing whitespace and non-printable characters; confirmed no text formatting inconsistencies across 1,200 rows.	Resolved
CR006	Checked for full-row duplicates by grouping on all 14 columns (Group By → Count Rows) in a dedicated reference query (Orders_Validation).	Confirmed 0 fully duplicate records in the dataset, in addition to the OrderID-level duplicate check in CR004.	Resolved

Verification Gate (Project 1 Threshold)

Requirement	Target	Result	Status
Error rate on unique identifiers (OrderID)	0%	0% (0 of 1,200 duplicated)	Pass
Error rate on date formats	0%	0% (all 1,200 rows in ISO 8601)	Pass

Methodology

All transformations were performed in Microsoft Excel using Power Query, preserving an untouched copy of the raw dataset (query: Orders_Raw) for audit purposes. Each cleaning action was captured as a named, ordered step in the Applied Steps pane of the working query (Orders_Clean): Set Initial Data Types → Rounded Prices to 2 Decimals → Filled Missing CouponCode → Converted Date to Date-Only → Trimmed and Cleaned Text Columns. Validation was performed in a separate reference query (Orders_Validation) that leaves Orders_Clean untouched: a Group By on OrderID confirmed the primary key (CR004), and a second Group By across all 14 columns confirmed zero full-row duplicates (CR006). Both results were cross-checked against row counts in the worksheet using ROWS and COUNTIF formulas.